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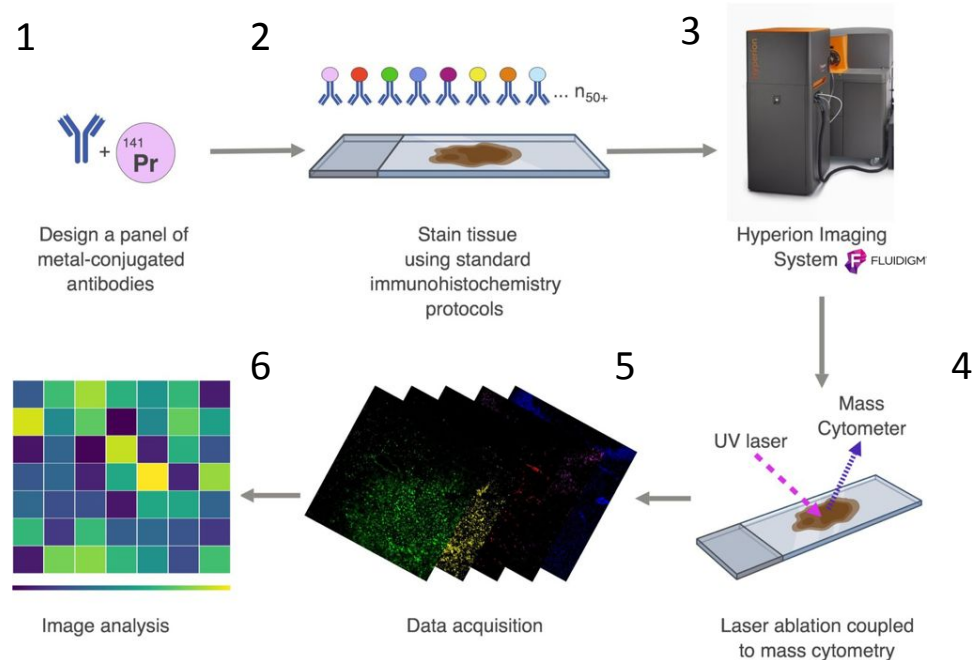
NSCLC Tumor IMC Analysis Updates

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What is Imaging Mass Cytometry (IMC)?

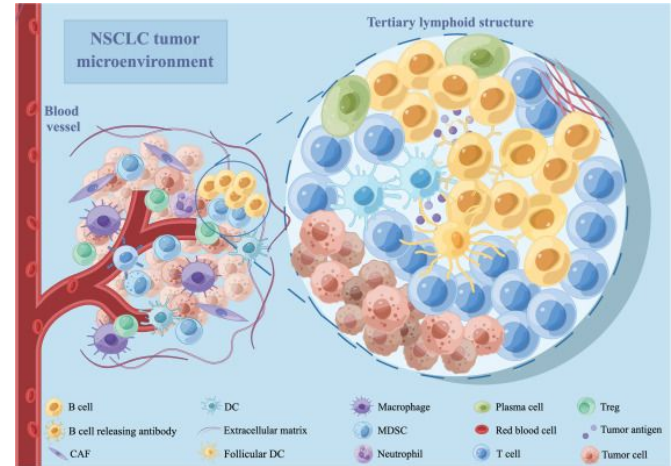
A spatial proteomics technique that can detect 40+ protein markers at single-cell resolution in tissue samples.

- Used to perform highly multiplexed imaging.
- Combines metal-labeled antibodies + laser ablation mass spectrometry



Why IMC for Non-small cell lung cancer (NSCLC)?

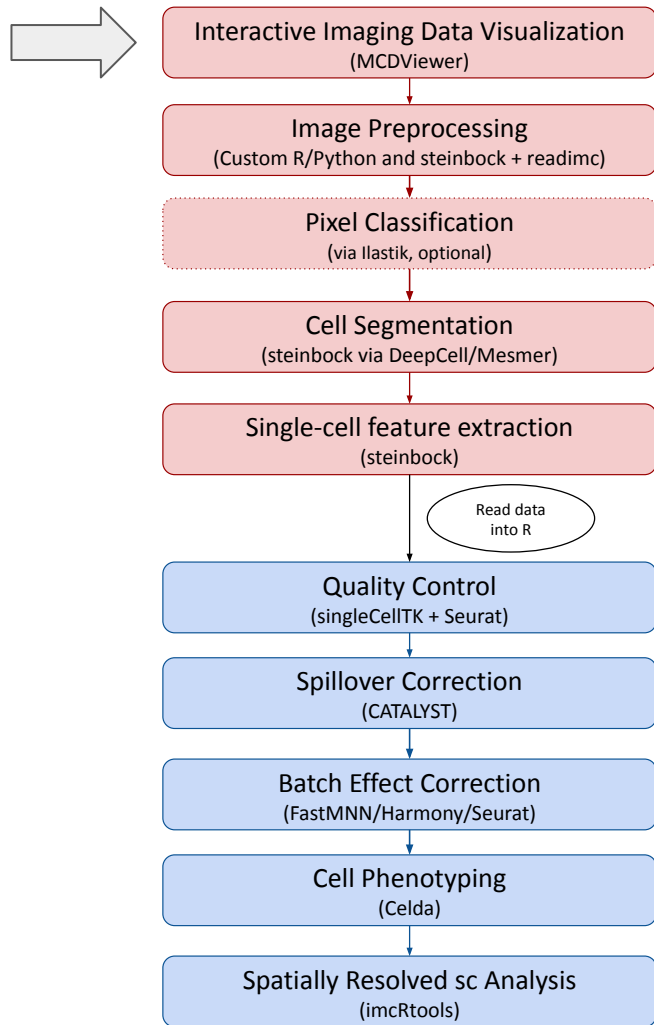
- Need to look at **tumor microenvironment (TME)**
 - Mix of tumor, immune, and stromal cells
 - Allows us to understand why treatment response varies so much across patients
- Traditional methods fall short
 - **Flow cytometry** → tells you what cells are present but not where
 - **Bulk RNA-seq** → averages signal across thousands of cells, loses spatial info
 - **Standard IHC** → keeps spatial context but limited to 1-5 markers at a time



IMC Data Analysis Workflow

* Spillover Correction requires another wet lab method so skip for our workflow

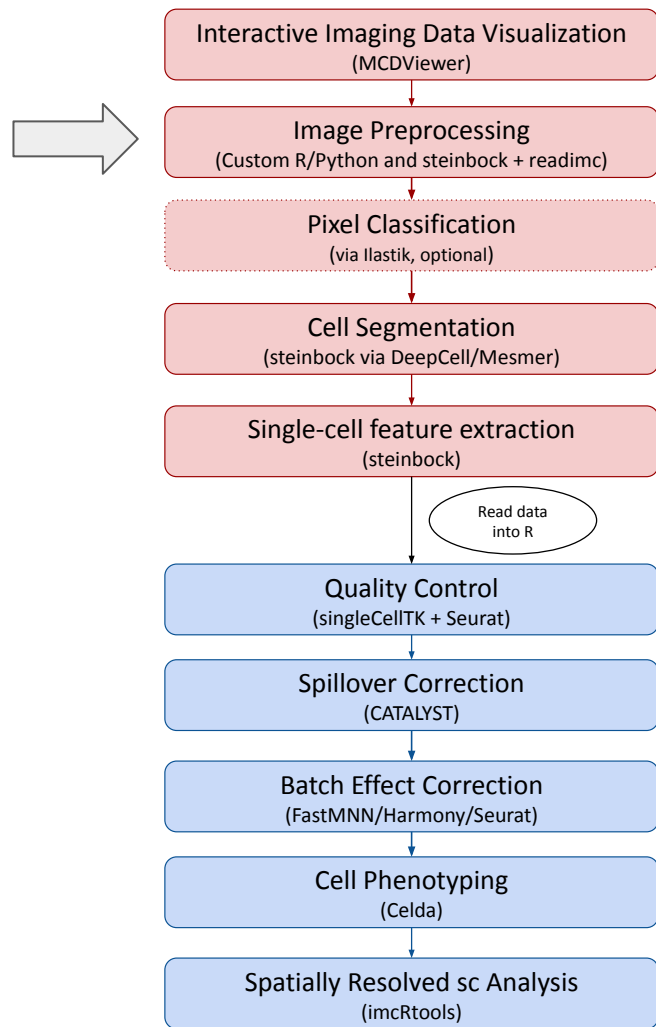
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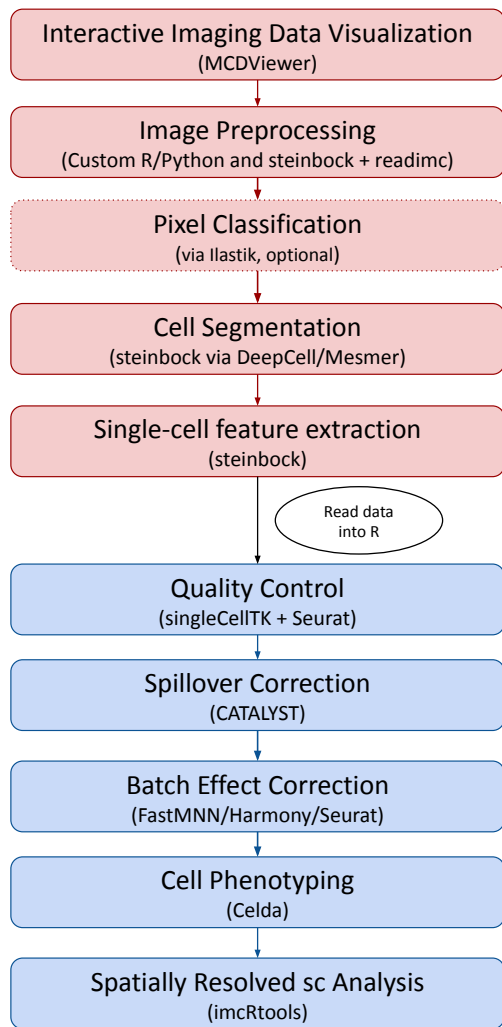
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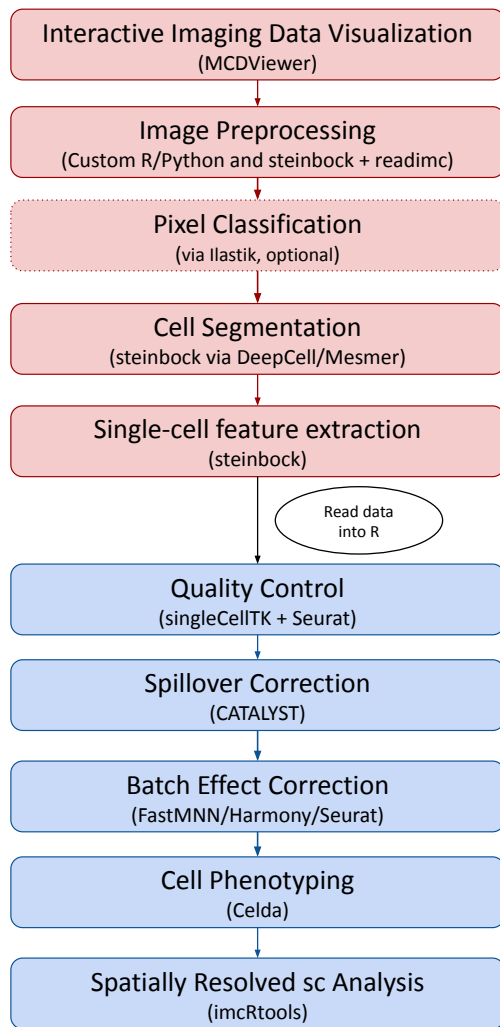
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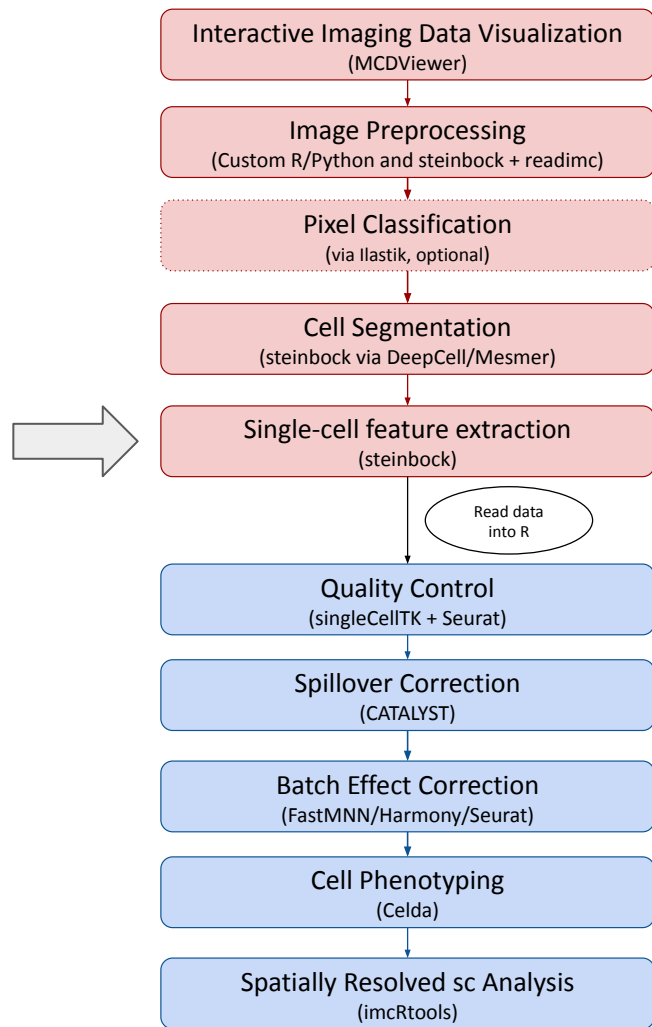
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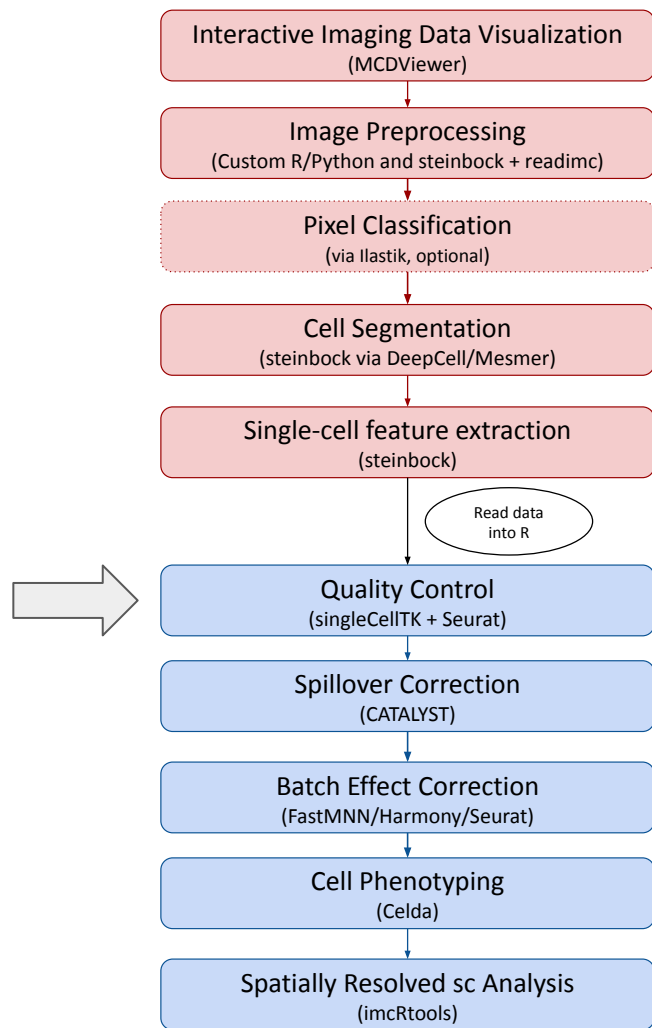
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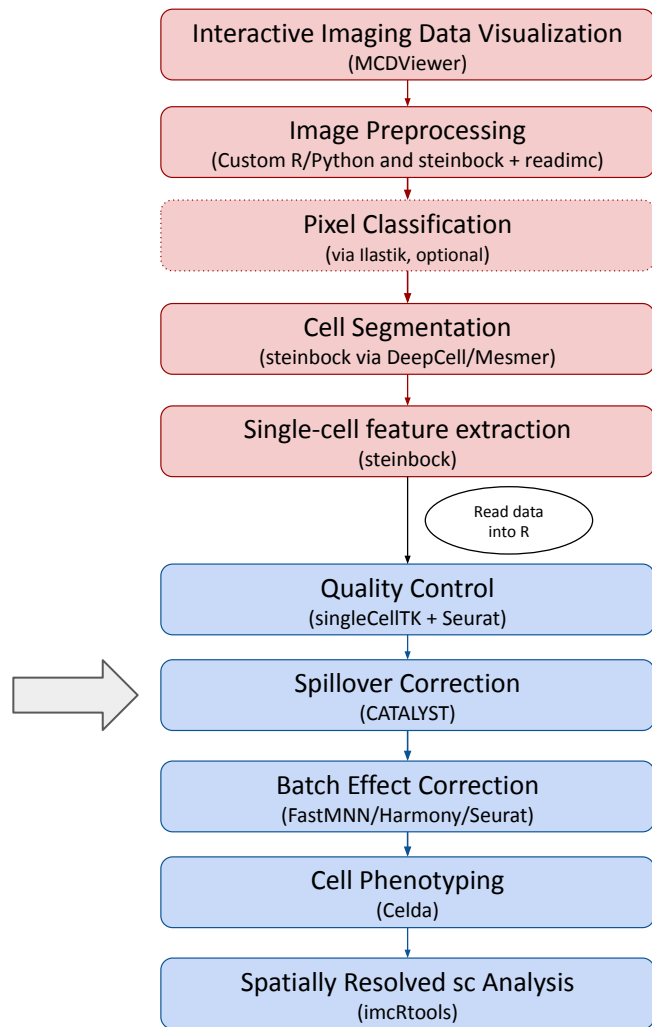
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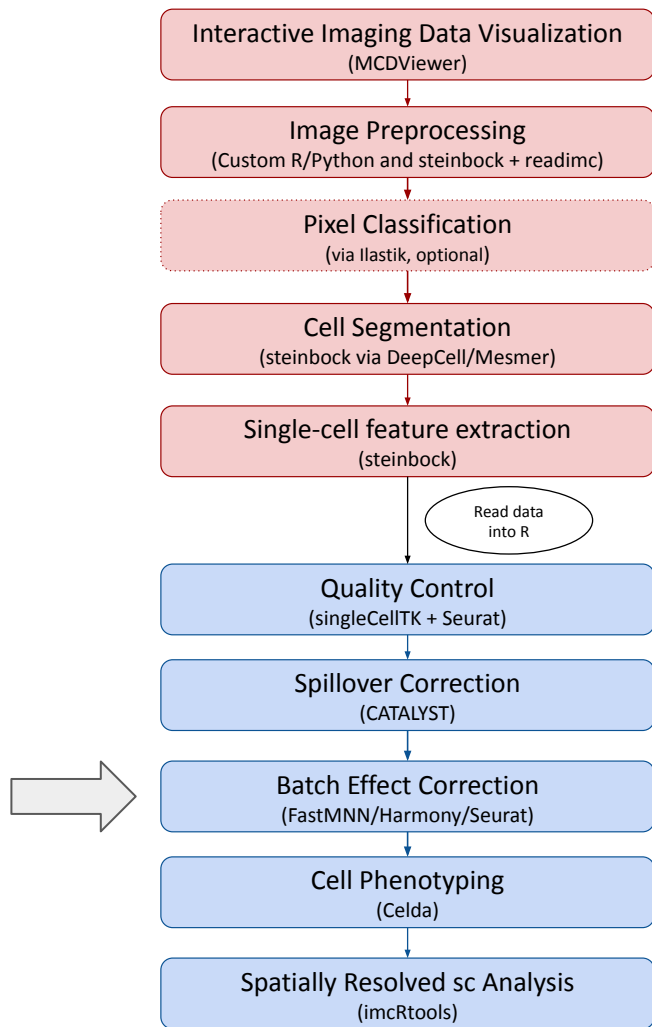
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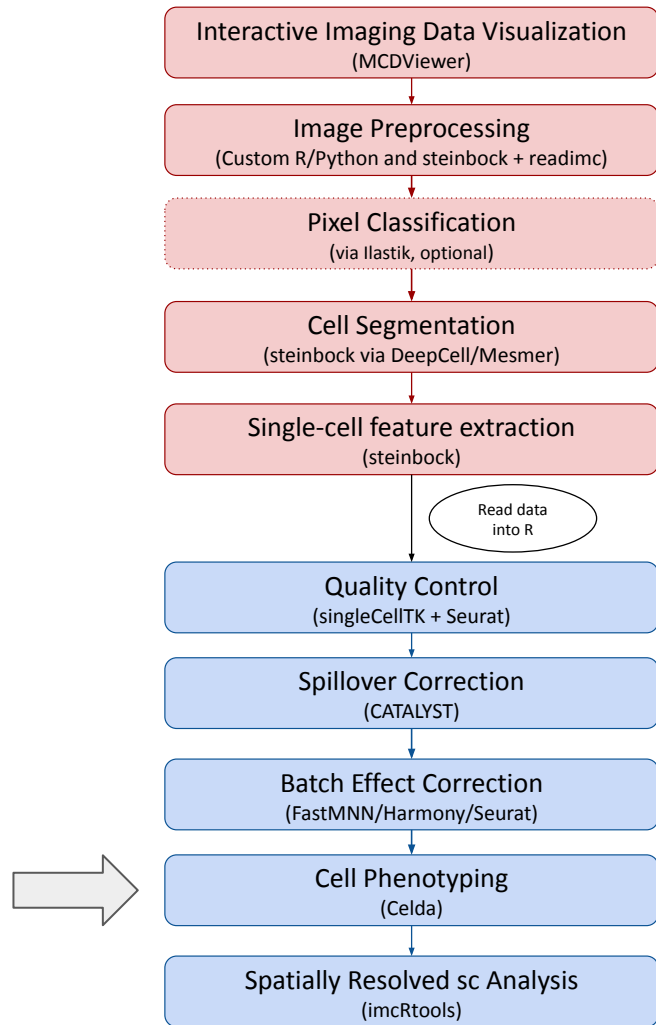
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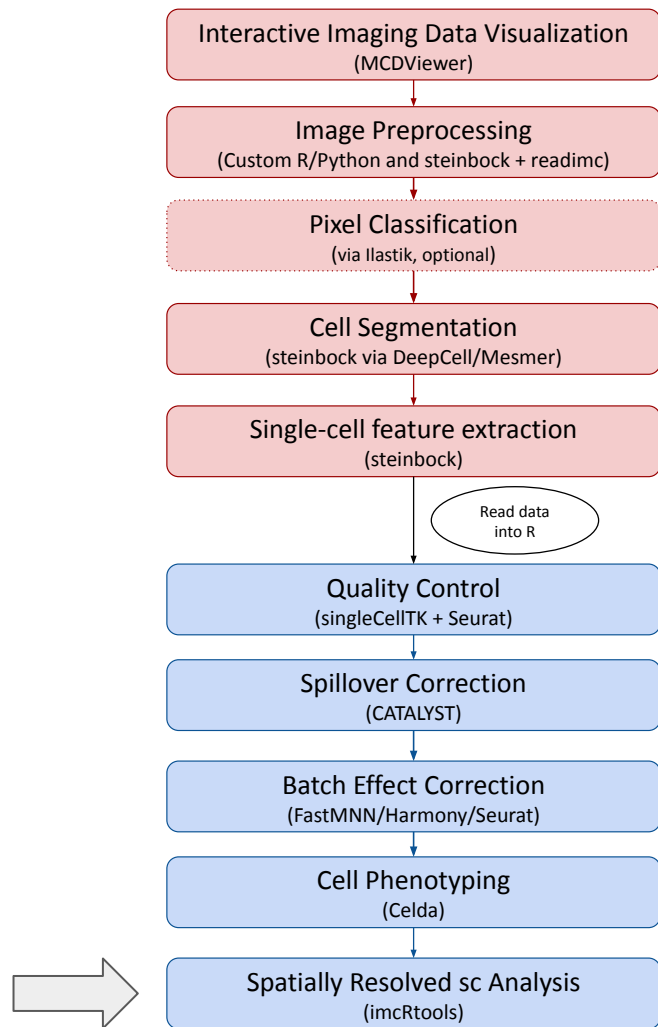
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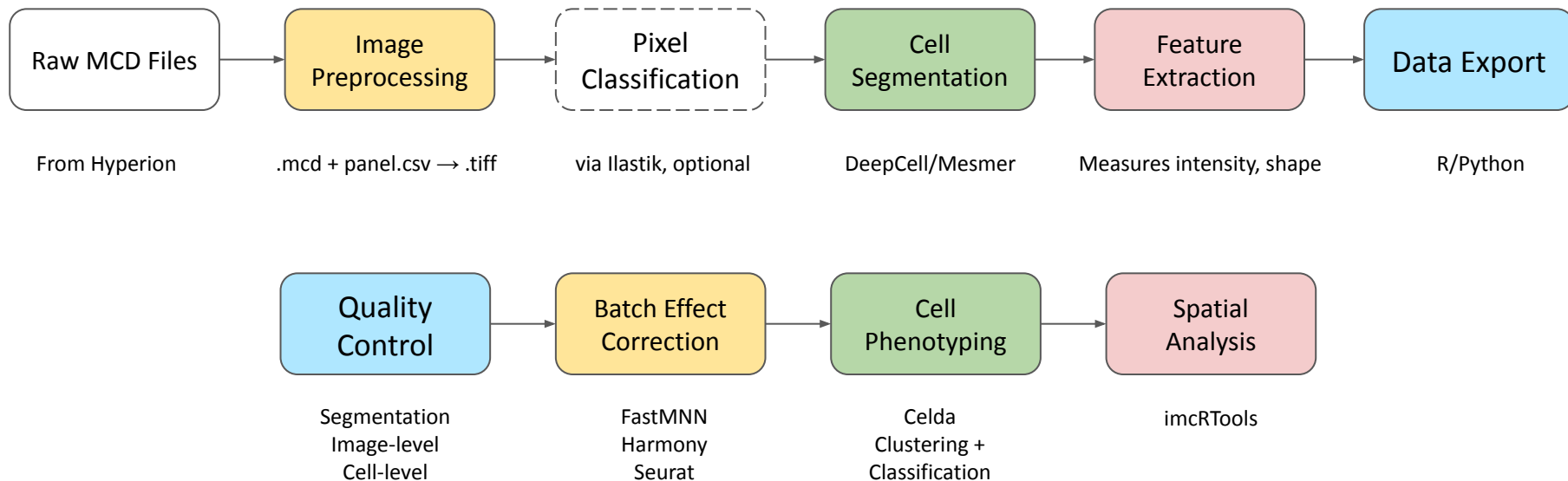
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IMC Data Analysis Workflow - An Overview

Steinbock: A computational toolkit for processing multiplexed tissue images from raw IMC data to single-cell features



Steinbock - Algorithmic Components

There are two algorithms for cell segmentation

- 1) Pixel classification via Ilastik + CellProfiler → requires pixel annotations
- 2) Deep learning via DeepCell/Mesmer

References

- **Analysis Workflow for IMC Data:** <https://bodenmillergroup.github.io/IMCDataAnalysis/index.html>
- **Steinbock Documentation:** <https://bodenmillergroup.github.io/steinbock/latest/>
- **Original IMC paper (Giesen et al, 2014):** <https://www.nature.com/articles/nmeth.2869>
- **Steinbock paper (Schapiro et al, 2021):** <https://pmc.ncbi.nlm.nih.gov/articles/PMC8916956/>
- **How Hyperion works:** <https://www.imc.unibe.ch/technologies/hyperion>